Assessing and Forecasting SO_2 Pollution in the Continental United States : A Data-Driven Approach to Mitigating Health Risks

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Introduction / Motivation

Air quality deterioration due to SO_2 emissions is a significant environmental issue that poses serious risks to public health.

It leads to a range of respiratory diseases, including asthma, bronchitis, and other chronic lung conditions. (especially vulnerable populations, such as children, the elderly, and individuals with pre-existing respiratory conditions)

The impact of SO_2 is exacerbated in densely populated urban areas, where exposure to harmful air pollutants occurs more frequently and intensively.



Research Questions

- How do meteorological and environmental factors influence SO_2 concentration levels in the U.S?
- How accurately can we predict future SO_2 concentration levels in different region of U.S using historical data and meteorological conditions?
- What is the estimated exposure of the U.S population to harmful levels of SO_2 and which populations are most at risk?

Literature Review

- The study addresses the concentrations of criteria pollutants in the contiguous U.S. from 1979 to 2015 and highlights the importance of prediction model parsimony in integrated empirical geographic regression models. -Giorgi, E. (2020).

- The research forecasts future SO_2 emissions, analyzing various emission scenarios and environmental factors. The study models the impact of SO_2 emissions on climate change, highlighting the potential effects of emission reductions. -Smith, S. J., Pitcher, H., & Wigley, T. M. L. (2005).

- The report comprehensively assesses the health impacts of SO_2 and its derivatives, establishing health criteria related to sulfur oxides. It emphasizes the relationship between SO_2 exposure and respiratory diseases. -U.S. EPA. Integrated Science Assessment for Sulfur Oxides - Health Criteria (2017).

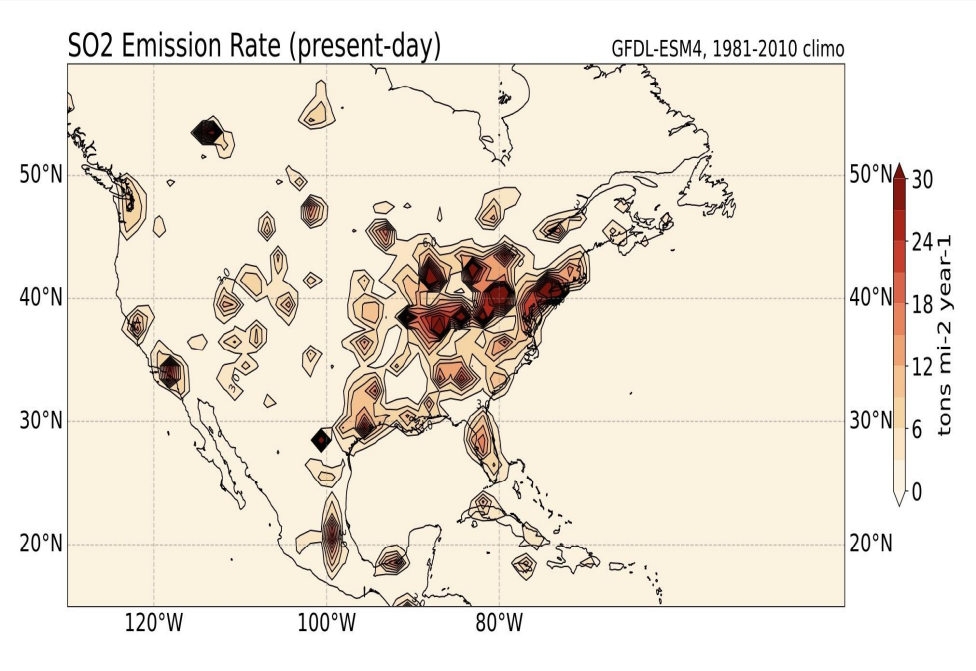
Climate Model

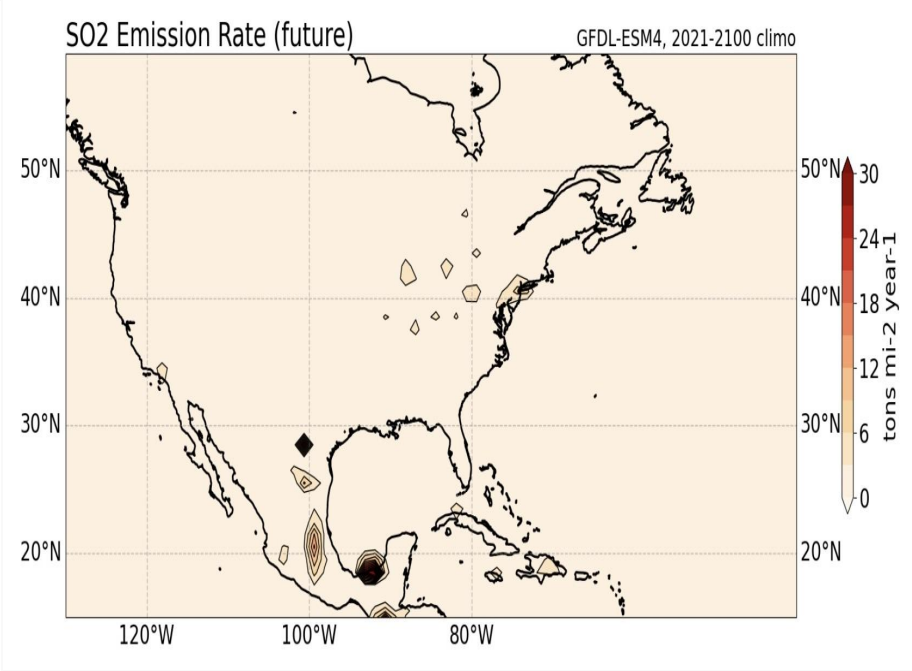
- SO_2 data
  - Model: GFDL-ESM4
  - Resolution: 1 x 1.25
  - Core: AM4
  - Experimental design: historical, SSP585

- Variable: Total Emission Rate of SO_2 (emiso2)

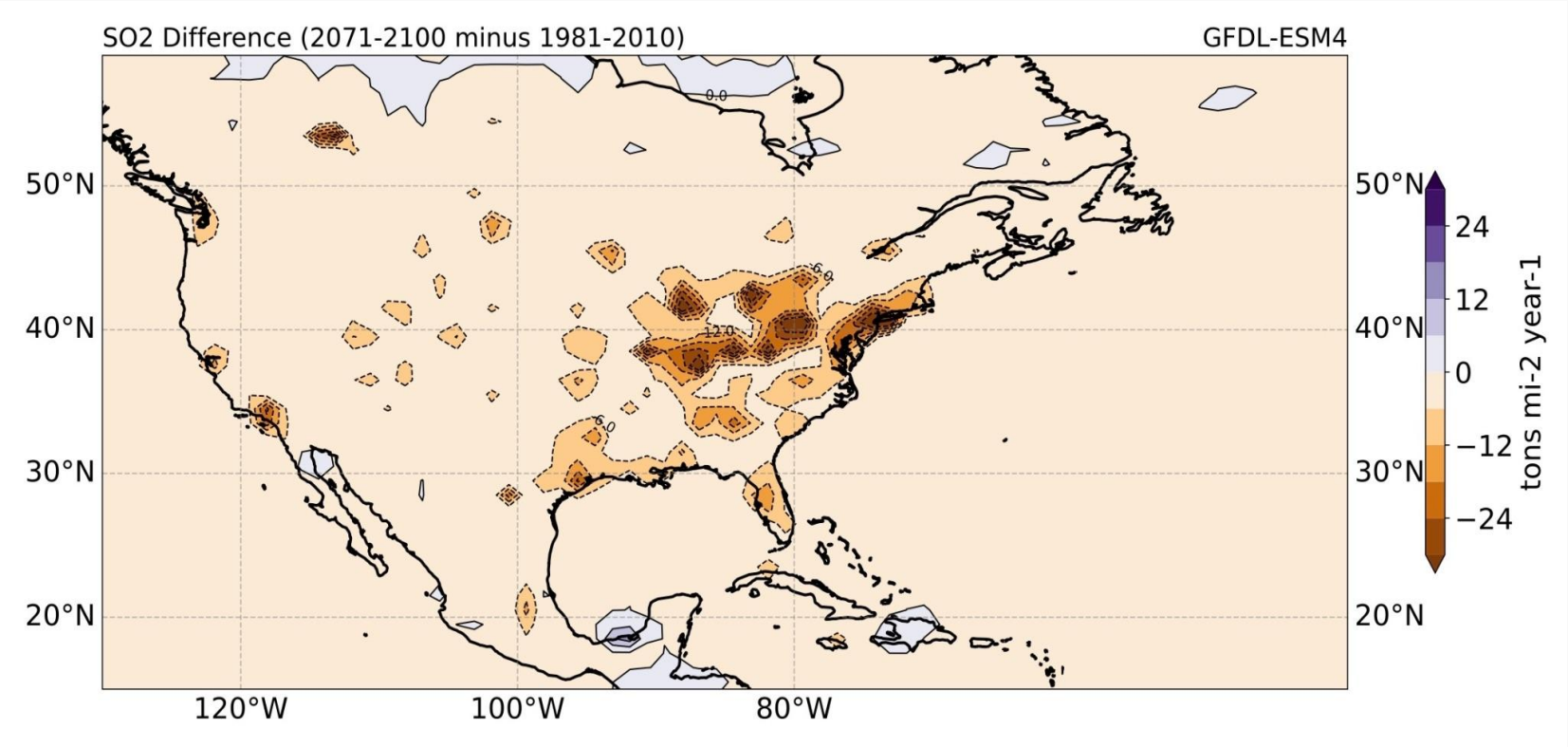
- Advantages and Limitations
  - Limitations: cannot account for future policies that may be implemented / relatively low resolution
  - Advantages: useful for understanding large-scale climate predictions and overall trends

Annual Mean of SO_2

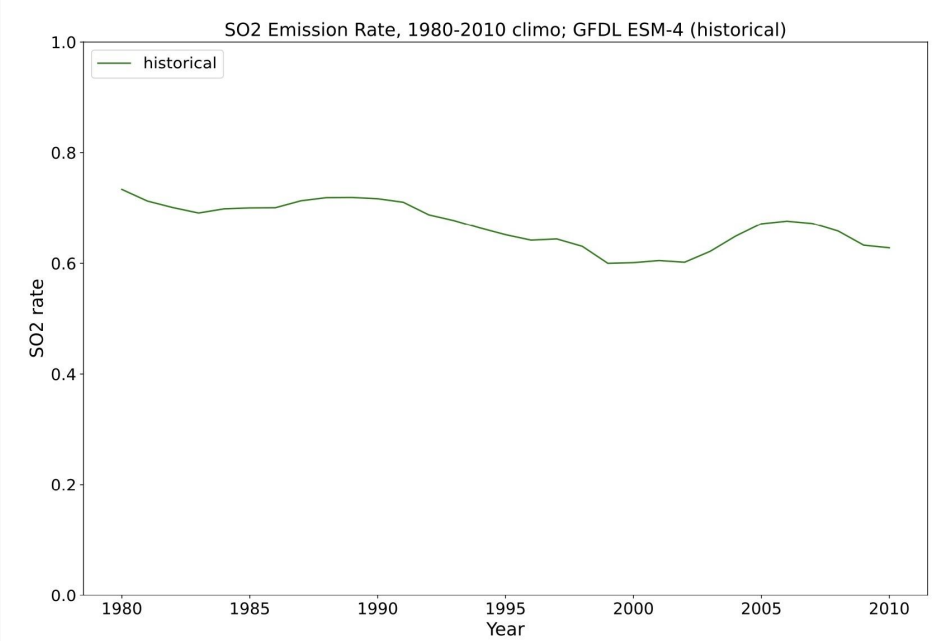


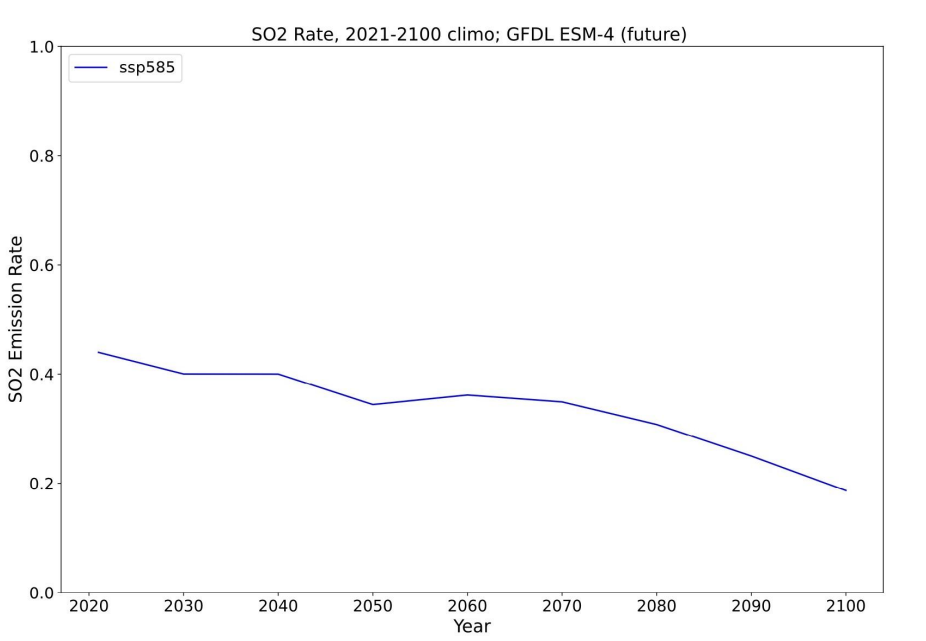


Change in SO_2 (Future vs Present)

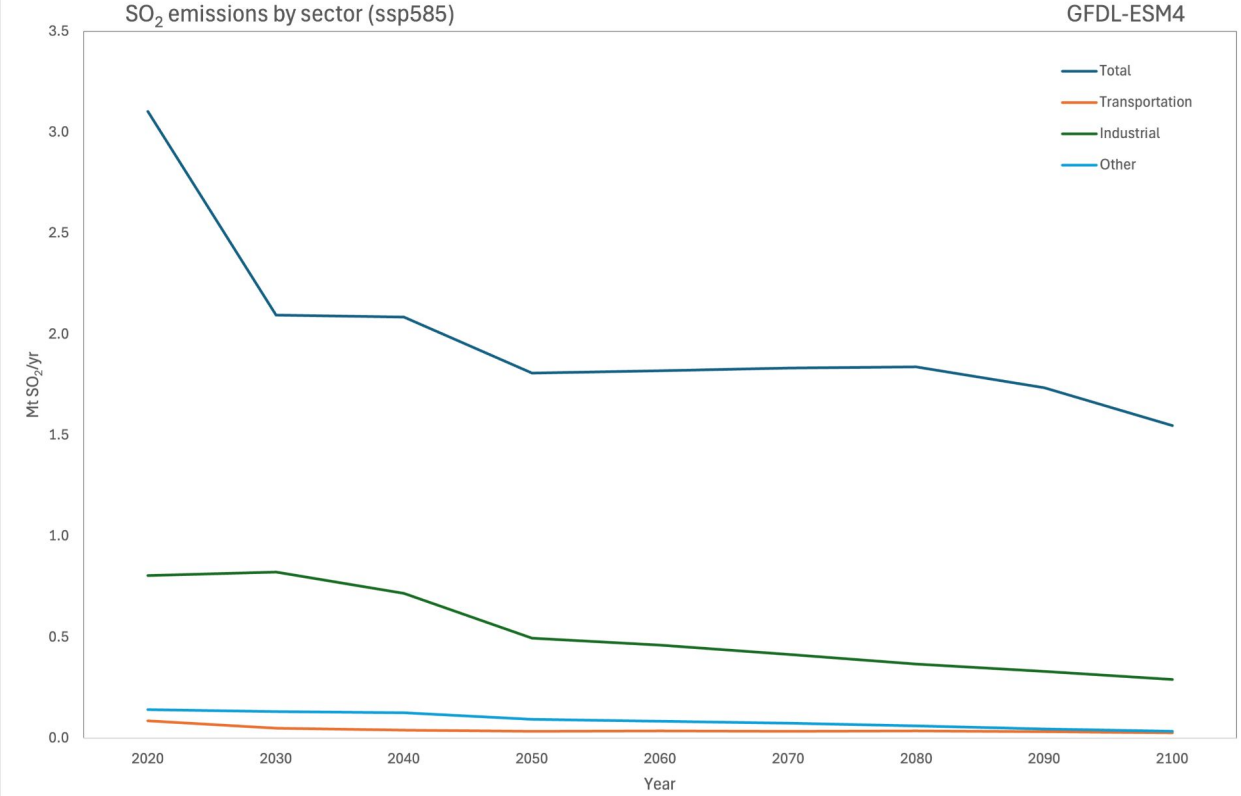


Annual Mean of SO_2 (Future vs Present)

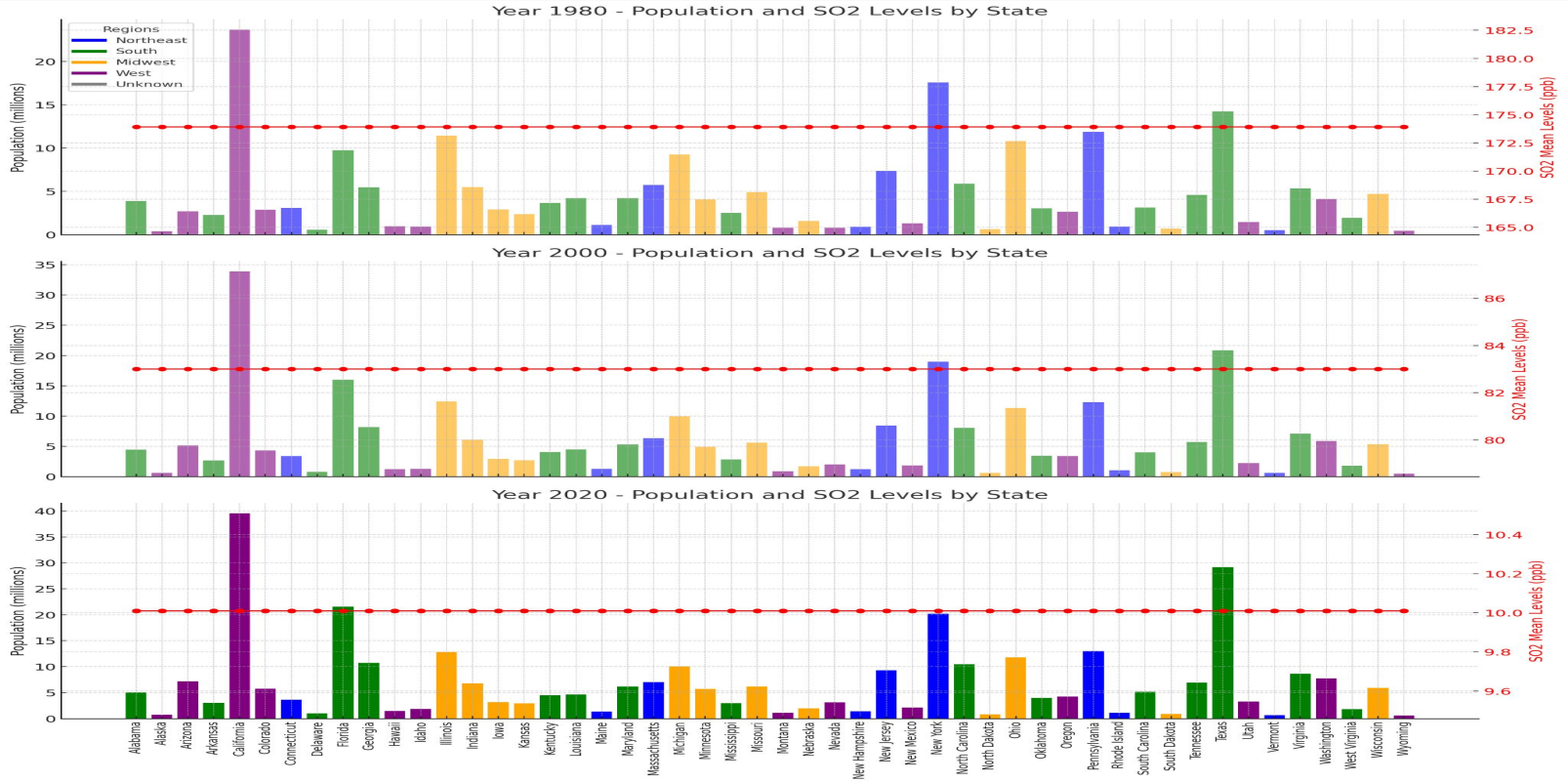




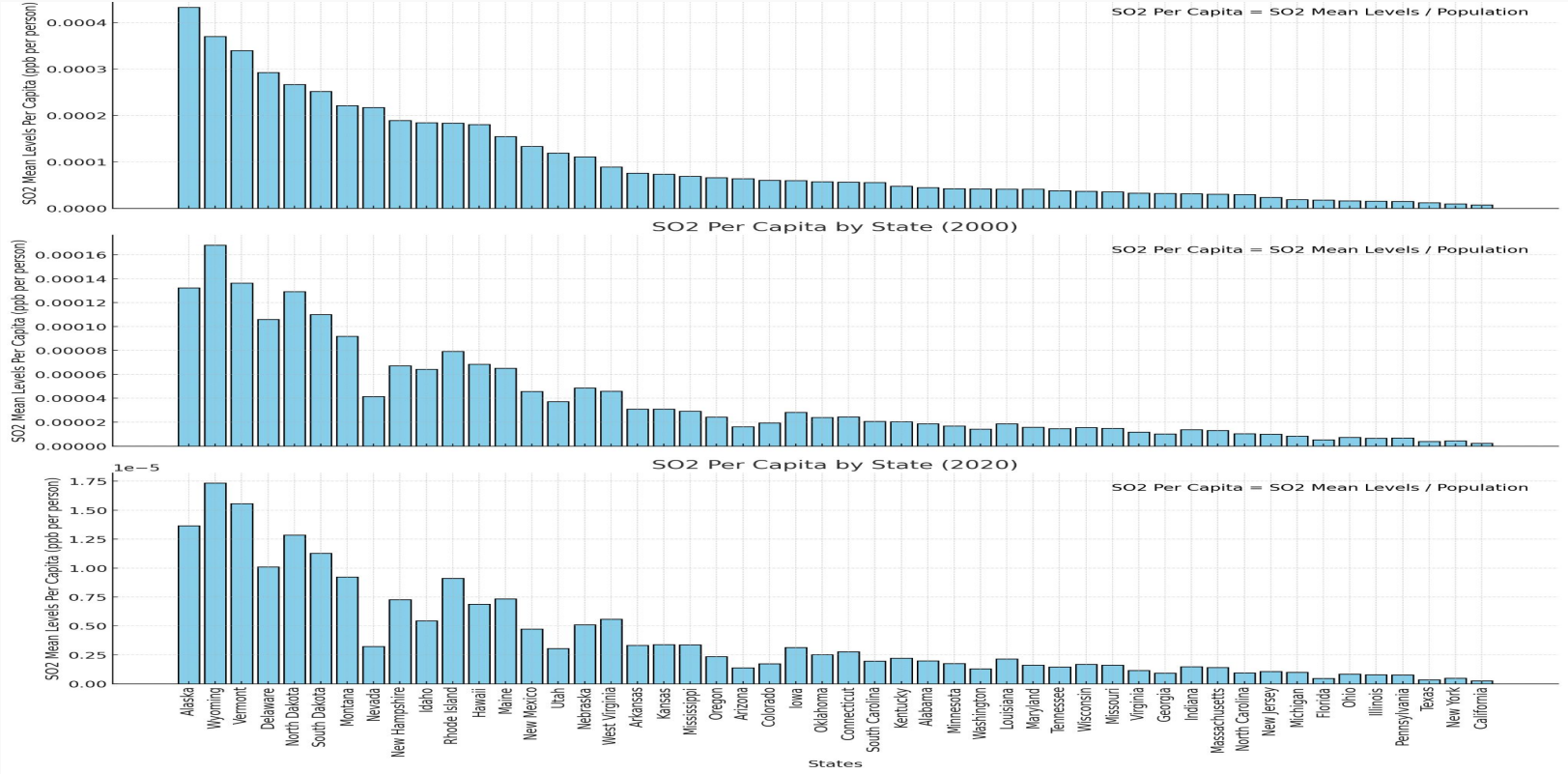
SO_2 emissions by sector (2020-2100)



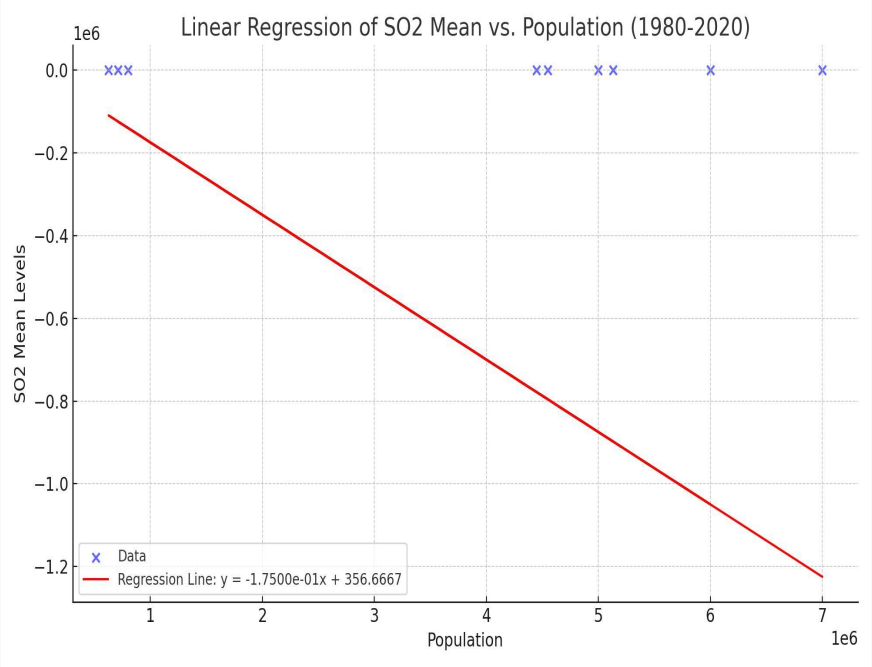
Population and SO_2 Levels by State (1980-2020)

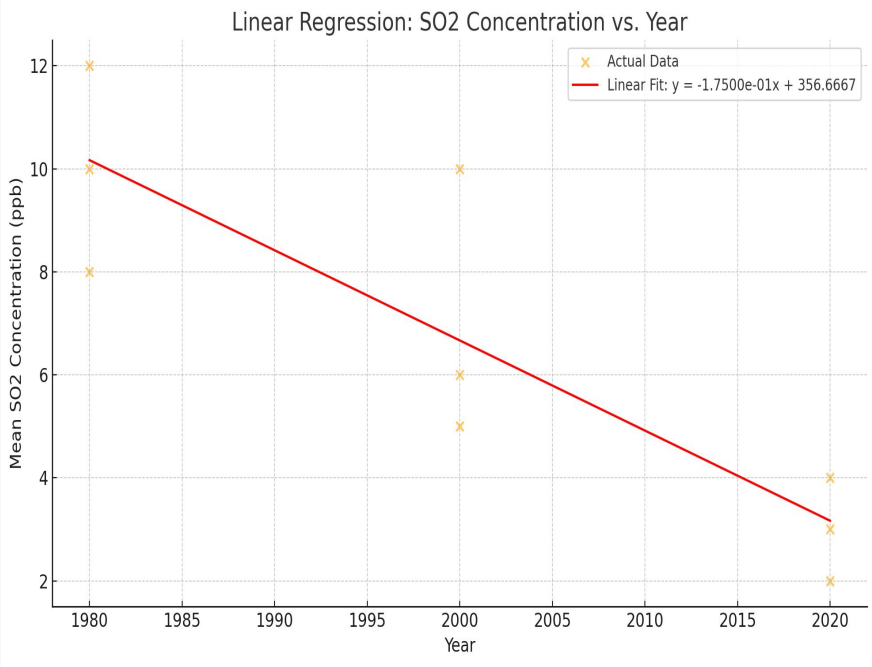


SO_2 Per Capita by State (1980-2020)



Linear Regression Models





Results

Regression Model Statistics:

Linear Regression: $y = mx + b$

y is the dependent variable (SO_2 Mean Levels), m is the slope of the regression line, and
x is the independent variable (Population), b is the y-intercept.

Linear Regression for SO_2 (1980): $y = 4.4823 \times 10^{-21} x + 173.9103$
Linear Regression for SO_2 (2000): $y = 2.6044 \times 10^{-21} x + 83.0115$
Linear Regression for SO_2 (2020): $y = 4.8464 \times 10^{-22} x + 10.0103$

Correlation Coefficient > 0

Conclusions (Population vs SO_2)

- Significance of Results:

Population vs SO_2 pollution: Linear Regression analysis shows that population growth has no significant effect on SO_2 emissions. The correlation coefficients for the two variables were less than .01. Showing little evidence of relation.

Population Size vs Pollution Per Person: SO_2 per capita analysis shows that populous states have lower SO_2 per capita despite larger population, which could more than likely be correlated with the states environmental policies.

- Limitations:

Technological and Industrial Changes: Over time, technological advancements or shifts in **industrial activity** (e.g., a shift from coal to natural gas) may influence **SO_2 emissions**, and these changes may not correlate with population trends in a straightforward way.

- Future work:

Consideration of Economic, Technological, and Policy Factors:

Objective: Future research should incorporate additional factors such as **economic growth**, **technological advancements**, and **policy changes** (e.g., the **Clean Air Act** or state-level emissions regulations) that significantly impact **SO_2 emissions**.

Action: Develop a **causal model** to explore how these external factors, beyond population growth, contribute to SO_2 emissions and other pollutants.

Member Contributions

Jenna Lee:

- Analysis of SO_2 data
- Task Manager and Scribe

Daniel McMoore:

- Analysis of SO_2 & U.S population data
- Linear regression calculation

References

1) Giorgi, E. (2020). Concentrations of criteria pollutants in the contiguous U.S., 1979 – 2015: Role of prediction model parsimony in integrated empirical geographic regression. *PloS One.*, *15*(2). https://doi.org/10.1371/journal.pone.0228535

2) SMITH, S. J., PITCHER, H., & WIGLEY, T. M. L. (2005). Future sulfur dioxide emissions. *Climatic Change*, *73*(3), 267–318. https://doi.org/10.1007/s10584-005-6887-y

3) U.S. EPA. Integrated Science Assessment for Sulfur Oxides - Health Criteria (Final Report, December 2017). U.S. Environmental Protection Agency, Washington, DC, EPA/600/R-17/451, 2017.

4) U.S. EPA. Policy Assessment for the Review of the Primary NAAQS for Sulfur Oxides (Final Report, May 2018). U.S.Environmental Protection Agency, Research Triangle Park, NC, EPA-452/R-18-002, 2018.

Thank you for listening!